2024 IEEE 25th Latin American Test Symposium (LATS) | 979-8-3503-6555-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/LATS62223.2024.10534623

Silicon Lifecycle Management Based on On-Chip Cross-Layer Sensing and Analytics for Space Applications

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Invited Paper

***Abstract*— IHP is one of the world's leading research institutions in the field of silicon/germanium electronics. In this field, it has extensive, closely coordinated expertise in semiconductor technology, materials research, high-frequency circuit design and system solutions that are currently applied in several domains, but in particular in the space one. Space domain, across the years, has become the natural IHP's vocation. This paper briefs the recent technological achievements and pave the next years' roadmap of the IHP's System Architectures (SyA) Department on the design of cutting-edge integrated circuits (ICs) for space application. Among these achievements, it is worth mentioning the development of multicore processors based on the RISC-V architecture, silicon lifecycle management (SLM) based on on-chip sensors and analytics, and an open-access reliability framework to allow ease-access to the company technology.**

***Index Terms*— RISC-V, on-chip sensor, on-chip analytics, silicon lifecycle management (SLM), in-field cross-layer information, space system.**

I. Introduction

Through its research and manufacturing services, IHP has significantly contributed to the innovative strength of Germany and Europe, especially in the field of ultra-high-frequency electronics and space domains. The institute has a pilot line that manufactures ICs using its high-performance mid-range SiGe BiCMOS 250 and 130 nm (conventional and radiation-hardened) technologies. In the speed of silicon-based transistors, IHP holds the world record with 720 GHz maximum oscillation frequency. In the space domain, IHP has concluded the evaluation process to include the 250 nm technology in the European Preferred Parts List (EPPL) of the European Space Agency (ESA) and is concluding similar procedure for the inclusion of its 130 nm radiation-hardened technology into the same list.

In this scenario, the IHP's System Architectures (SyA) department is conducting research in four main topics: (1) Wireless broadband systems, (2) Fault-tolerant computing, (3) Design and test methods for digital circuits, and (4) Hardware security. Despite the obvious importance of the other topics and the vast IHP contributions in these domains along the recent years, for the sake of space and conciseness, this paper is focused only on topics (2) and (3).

In the field of fault-tolerant computing systems for demanding applications such as space or automated driving, the IHP's major focus is on dependable multi-processing architectures (mostly based on the RISC-V open-source instruction set architecture – ISA running a customized version of the FreeRTOS operating system). For these applications, we are developing strategies to dynamically trade-off performance, power consumption and reliability in the field. In this context, the rad-hard design methodology is paramount, since space is one of the strategic fields of application for IHP. The SyA department also actively pursues current research topics on artificial intelligence (AI), electro-photonic interfaces, neuromorphic computing, and hardware architecture oriented for on-chip sensing and analytics.

Recently, IHP has been devoted to open design and open tools towards establishing an OpenPDK program. The final goal is to construct an open-access reliability framework to allow IC designers from all over the world to access, easily and free-of-charge, not only the conventional 130 nm SiGe BiCMOS technology, but also its rad-hard one. So far, the open-access library counts not only with almost one hundred basic standard cells, crucial bricks in any IC design process, but also with more elaborated cells such as SRAM blocks and sensors.

The remaining of this paper is organized as follows: Section 2 describes the IHP's solutions for RISC-V-oriented on-chip sensing and machine learning-based analytics. Section 3 describes the solutions IHP is paving towards silicon lifecycle management (SLM) to design resilient systems for space applications. Finally, the last section tailors the conclusions of the paper.

II. IHP Solutions

II.1 On-chip sensing

IHP is investigating various on-chip sensors for monitoring system's state and collecting information on transient and permanent faults. In the following, the most significant solutions are presented.

A. Silicon-level information

(i) Multi-purpose reliability sensor:

a) Monitor aging of delay-critical paths of combinational logic. As technology scales down, aging is becoming one of the biggest concerns when designing ICs not only for space applications, but also at the terrestrial level. This phenomenon can be explained mainly due to degradation mechanisms such as hot carrier injection (HCI) and bias-dependent temperature instability (BTI) that change the transistor threshold voltage (V_{th}) as defined during the manufacturing process. It is worth noting that this sensor can also be used to improve the device

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screening process, during production test. This sensor is an evolution of the aging sensor first proposed in [1] and [2].

b) Detect single-event upset (SEU) in flip-flops (FFs) and single-event transient (SET) in combinational logic. As technology scales down, bit-flips became one of the biggest concerns when designing ICs for space applications. This is due basically because of the reduction of the circuit internal node capacitances (which in turn reduces the sensitive volume used to store the information in memory elements such as FFs and RAM cells) and reduction of power supply voltage.

c) Monitor electromigration evolution in metal lines. Electromigration became a progressively severe reliability challenge due to the always-increasing interconnect current densities that can be measured in nanoscale ICs.

Currently, IHP is integrating multi-purpose reliability sensors in a RISC-V processor. These sensors are distributed in the floating-point multiplier (FPMul), and in the L1 memory (Cache). In both cases, the sensors replace the output FFs connecting these functional blocks to the processor bus, and deal to monitor: (a) aging effects on critical paths of the FPMul and on the response time of the Cache, and (b) detect SEU and SET at the output FFs. The sensors also monitor the delay increase provoked by some cases of electromigration on Cache bit lines and on FPMul tracks. The sensors have been located in these specific blocks because they are among the most timing sensitive nodes of the processor, i.e., the nodes operating with the narrowest clock time slack. So, the nodes with the highest probability to fail as aging evolves during chip in-field operation. The RISC-V processor is being designed with the IHP 130 nm SiGe BiCMOS rad-hard technology.

In another activity, IHP is coupling the multi-purpose reliability sensor with a commercial 5nm (TSMC) SRAM IP core. Experimental results based on extensive SPICE simulations demonstrate the approach is able to detect slight delay increases in the memory time response due to aging, much earlier than a functional error can be observed at the memory output. This solution allows performing mission-mode monitoring of memory operation, which is a crucial aspect of silicon lifecycle management (SLM) approach.

(ii) Sensor for monitoring power-supply activity:

Noise on power-supply lines causes voltage level bounce that, in turn, produces noise and jitter in the system and ultimately, bit-flips in memory elements. In [3], authors developed work to use Artificial Neural Network (ANN) to perform in-field identification and classification of different types of noise conducted in the IC DC power lines according to a specific set of IEC standards. After identification, proactive actions can be taken to guarantee the expected IC robustness. The next steps of this work are: (a) development of on-chip sensor to monitor power-supply bus activity, and (b) development of a machine learning model that, based on the classified type of noise conducted in the IC DC power lines as above mentioned, takes the appropriate action to optimize reliability accordingly. For instance, slow down IC clock frequency, increase power supply voltage and/or activate error detection & correction (EDAC) functions during the period the system is operating under such noise exposition.

(iii) Sensor for monitoring temperature across the die:

IHP has been investigating self-checking thermal sensors based on two ring-oscillators and duplication with comparison for critical digital blocks. If a fault occurs in one sensor, external logic can reset it or activate a redundant one. This is an interesting solution, considering that ring oscillators are sensitive to aging, and processing logic may be also sensitive to soft errors, potentially resulting in jeopardized measurements.

(iv) Sensor for monitoring aging of SRAM cells:

Note that the sensor presented at the end of section *i(a)* is able to monitor the increase of the cache response time due to aging of the memory block as a whole. However, it is not able to discriminate the aging status of individual internal structures such as the memory cell and peripheral logic (read/write logic, line/column decoders, hang-up logic and sense amplifiers). It is worth noting that the most aging-sensitive structure in a memory block is the memory cell, whose transistor sizing must be precisely adjusted during the design phase to guarantee the necessary cell stability during write and read operations, in the shortest possible time. Also, the cell must be able to operate correctly even under stress conditions (e.g., noise on power supply lines, low-voltage operation, process/temperature variations and wear-out). Therefore, we are analyzing the possibility of migrating the on-chip sensors first presented in [4] to our 130 nm rad-hard SiGe BiCMOS technology. Once advanced cell aging state is detected, rejuvenation algorithm can be applied to the cells (or block of cells) according to [5] in order to recover the memory from aged states.

(v) Glitch sensor and suppressor for data and control lines:

Besides noise across the power supply lines, the noise on data and control lines (e.g., reset and clock) may be also caused by electromagnetic interference generated at the board level and propagated into the chip as conducted noise. This noise may also cause soft errors. Thus, dedicated sensors may also be needed to detect and counteract such soft errors in critical combinational circuits. In [6], we have proposed a simple circuit acting as a glitch filter and detector. This circuit filters out glitches shorter than the predefined threshold, and uses registers to record the number of detected glitches.

B. System-level information

In addition to the silicon-level information (aging, SEU, SET, power-supply and temperature) collected as above mentioned, there are also reliability sensors being designed to monitor the operating system activity. The goal of these sensors is to detect faults that escape detection by the native fault detection mechanisms embedded in the kernel of the operating system, in particular, those faults that: (a) affect the task scheduling process, (b) violate task execution deadline, and (c) violate time and space allocated by the operating system for every task in a mixed-critical execution environment. The sensors are being designed to monitor *static* scheduling algorithms such as Earliest Deadline First (EDF) and Rate Monotonic (RM), as well as more advanced scheduling algorithms based on *dynamic* priority allocation. These algorithms are executed under the control of a customized version of the FreeRTOS kernel. The primary goal of detecting fault types as defined in (a), (b) and (c) is to support compliance with ARINC 653 standard [7], which specifies an operational environment for application software used in Integrated

Modular Avionics (IMA) modules. It allows the simultaneous execution of programs with mixed degree of criticality on the same computer platform. To detect these types of faults, the sensors must be configured, at least, with the following information for each one of the tasks to be executed by the processor: worst-case execution time (WCET), deadline, periodicity, allocated memory space, priority, processor core allowed to execute each task, and access permissions (for read and/or write). These sensors are inspired by those first published in [8] and [9].

C. High-energy particle counter and screening

The soft error rate of an IC may vary considerably due to varying operating conditions. For example, in space radiation environment, the soft error rate may increase several orders of magnitude due to solar particle events or due to the South Atlantic Anomaly (SAA) region, near the Equator line. In such conditions it is desirable to employ dedicated on-chip infrastructure not only to *detect* and *correct* bit-flips in memory elements, but also to *count* the number of high-energetic particles striking the surface of the die that could provoke an SEU. One of the most common particle counters is based on SRAM memory array. Such an array is coupled with a dedicated EDAC logic to detect, correct and count the number of SEUs is occurring in the memory per time interval. For this purpose, any SRAM can be used to determine the particle count rate, where the number of detected upsets per time unit can be correlated with the particle flux. IHP has proposed the use of embedded SRAM as a particle counter [10]. The advantage of embedded SRAM is that an existing on-chip memory is used both for data storage and radiation detection, thus minimizing overhead. This approach was implemented in the TETRISC quadcore processor and coupled with an AI model to hourly predict SEU rate in spaceborne electronics, as described in section II.2 [11].

Another solution to implement particle counters that IHP has investigated is the use of pulse stretching (skew-sized) inverters for monitoring the variation of count rate and linear energy transfer (LET) of energetic particles [12]. In this case, the particle counter is implemented as a cascade of two pulse stretching inverters, and the required sensing area is obtained by connecting up to 12 two-inverter cells in parallel and employing the required number of parallel arrays. The incident particles are detected as single-event transients (SETs). The SET count rate denotes the particle count rate, while the SET pulse width distribution depicts the LET variations. The advantage of the proposed solution is the possibility to sense the LET variations using fully digital processing logic. SPICE simulations conducted on IHP's 130 nm CMOS technology have shown that the SET pulse width varies by approximately 550 ps with a LET range from 1 to 100 $\mathrm{MeV \cdot cm^2 \cdot mg^{-1}}$.

However, soft errors may also occur in an environment where ionizing radiation is not present. For example, soft errors may be caused by electromagnetic interference that can be generated at the board level and propagate into the chip as conducted noise on power supply lines, data lines or control lines (e.g., clock). Thus, dedicated sensors may also be needed to detect and counteract such soft errors in critical circuits. In [11], we have proposed a simple circuit acting as a SET filter and detector. This circuit filters out any SET shorter than a predefined threshold, and uses registers to record the number of SETs that have been detected.

II.2 On-chip Analytics

A. TETRISC processor architecture

The aerospace industry's growing need for real-time data processing has emphasized the importance of multi- and manycore systems. In most of these systems, existing redundancy can be leveraged to dynamically adjust performance and fault tolerance as per evolving demands. Extensive research at the processor core level has yielded significant advancements in this area, such as the TETRISC (TETra Core System) SoC [13], [14] developed by IHP. This processor is based on the RISC-V open-source instruction set architecture (ISA), which in turn, is an adaptable and fault-tolerant quad-core version upgraded from the Pulpissimo single-core platform [15] (details in Fig. 1).

The TETRISC SoC hosts three types of on-chip sensors to monitor SEU, aging and temperature. Therefore, it is designed to respond to various environmental or intrinsic factors like temperature, radiation, and aging through its adaptive capabilities. Typically, the four cores operate independently at maximum power in high-performance mode. However, if safety-critical events are detected by the sensors, the system can adapt accordingly. For instance, if temperature in one of the four cores rises excessively, it can be individually switched-off to a destress mode using clock gating to facilitate cooling. In scenarios where an increased error rate occurs due to aging or radiation effects, the system can combine two, three, or all four cores into a lockstep configuration, executing tasks redundantly and systematically comparing the outputs. Additionally, if specific cores consistently process data inaccurately, they can be identified, deactivated, and the system can continue functioning (after task migration among the cores) at a degraded level to ensure operational continuity.

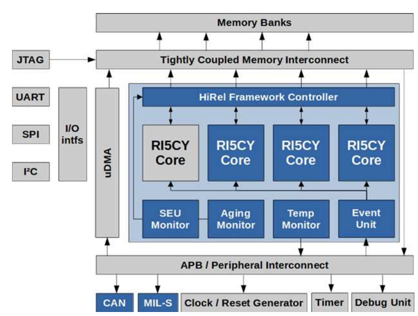

Fig. 1. Schematic overview on the TETRISC SoC.

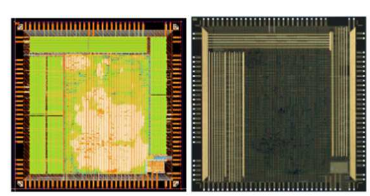

Fig. 2 Layout and die photo of the TETRISC SoC.

To comprehensively assess the system capabilities in terms of adaptability and fault-tolerance, we implemented the TETRISC architecture using the IHP's SG13S technology (130 nm node). Fig. 2 depicts the final layout and a die photo. The prototype, occupying a chip area of 43.56 mm², successfully met all specified requirements. Work on an enhanced iteration is already underway, targeting improvements in terms of technology node and new, updated on-chip sensors' versions. Moreover, stronger focus is given on hardware support to allowing the processor to run under the control of an improved real-time operating system. This new processor version will enable us to delve deeper into the system's capabilities concerning performance, sensing, fault-tolerance and adaptability.

B. SEU rate prediction based on AI-analytics

In space applications, it is critical to address the reliability issues caused by complex radiation environments on electronic systems. Solar events, in particular, can dominate the space radiation environment, potentially causing radiation levels to escalate by five orders of magnitude within hours or days. Such rapid changes can lead to a marked increase in fault rates for electronic systems, posing considerable risks to their reliability. To solve this problem, IHP is employing AI for analyzing and predicting space radiation patterns. By applying AI to analyze these intricate radiation environments and predict SEU rates during in-field-operation, we can provide essential proactive action to protect spaceborne electronics. This approach has been developed in conjunction with the TETRISC processor architecture, described in the previous section.

Fig. 3 illustrates the block diagram of the proposed AI-based methodology for predicting in-flight fault rates [16]. This method synergizes real-time data analysis with predictive modeling, aiming to accurately forecast SRAM SEU rates. It encompasses two phases: an initial *offline phase* focusing on historical space radiation data analysis and model training, followed by an *online phase* dedicated to real-time error assessment and optimization. The core of this approach involves selecting and training a time-series based AI model to process extensive historical space radiation data. During the online phase, the trained model is then integrated in a hardware accelerator [11], [17]. Moreover, during this phase, a learning algorithm is also employed to refine predictions of the target SRAM SEU rates in real-time response to evolving space radiation conditions.

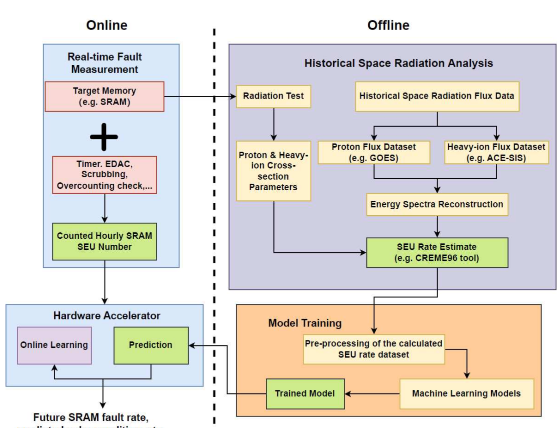

Fig. 3. Block diagram of the proposed AI method for in-flight SEU rate prediction.

In the *off-line* phase, we collected and analyzed data from 36 solar events during Solar Cycle 24. The raw historical space flux data are sourced from the ACE-SIS and GOES databases, which offered detailed space heavy-ion and proton fluxes data. By combining these data with cross-section information of the target SRAM, we were able to compute the SRAM SEU rate for every hour, during a period of 5,000-hour operation. In this work, five machine learning models were analyzed: (1) Linear Least Squares Regression, (2) Decision Tree, (3) K-Nearest Neighbors, (4) Multi-Layer Perceptron (MLP) Neural Network, and (5) Recurrent Neural Network (RNN) with Long Short-Term Memory (LSTM). It was observed that the RNN with LSTM model yielded the best results. The Linear Least Squares Regression performed almost equally good. By comparing these two models, the Linear Least Squares model presents the advantage that it is much simpler to be implemented in hardware, and so resulting in significantly less area overhead than the RNN/LSTM model. Therefore, the Linear Least Squares Regression model was the one implemented in the hardware accelerator.

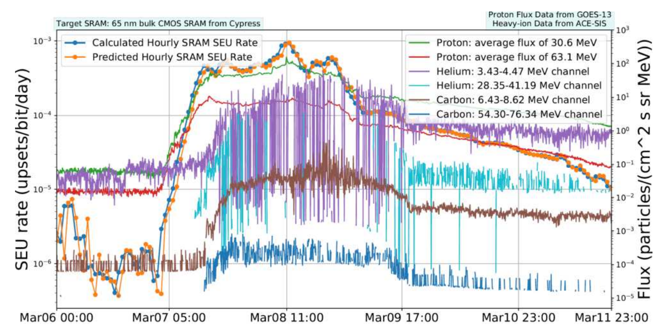

Fig. 4. Example of the AI-based SEU prediction method's performance for a 65 nm SRAM during the March 6-11, 2012 solar event.

Moreover, the hardware accelerator was further enhanced with an *on-line* learning method, the *Stochastic Gradient Descent (SGD)*, which enabled the system to dynamically adapt to the changing space environment. Thereby, enhancing the approach prediction performance in real-time. Consequently, by considering the counted hourly SRAM SEU number as provided by the Real-Time Fault Measurement unit, the accelerator can accurately predict the following hour SEU rate. For illustration, Fig. 4 displays the calculated and predicted particle-induced hourly SEU rate for a 65 nm SRAM module, as forecasted by our radiation predictor during the solar event that occurred from March 6 to 11, 2012.

III. Solutions for SLM

This section briefly describes the solutions that IHP proposes for the implementation of future intelligent robust systems for space applications. Silicon Lifecycle Management (SLM) is a relatively new process associated with the monitoring, analysis and optimization of semiconductor devices as they are designed, manufactured, tested and deployed in end user systems. SLM is based on two underlying principles: (1) gather as much useful data about each chip as possible, and (2) analyze that data throughout the chip's entire lifecycle to gain actionable insights to improve chip and system related activities. The SLM Platform to be developed will gather data from the chip lifecycle at three moments: (1) during *IC characterization*: to assess the device screening process based on reliability and wear-out tests; (2) *production test*: to assess the device screening process based on noise immunity, signal time slack in delay-critical paths, minimum operating voltage and I_{DDq}; and (3) *in-field operation*:

to optimize system towards user needs of lifespan extent, performance, power, reliability and security. To do so, the SLM Platform performs analytics on cross-layer system information. Fig. 5 depicts a general view of the proposed SLM Platform.

The outputs of all on-chip sensors distributed across the processor should be collected by a dedicated standard wrapper interface supporting AMBA and JTAG test access protocols, for instance. The interface goal is twofold: (a) format and compact all sensors' outputs into a single error indication (EI), which in turn drives an output pin of the processor, and (b) connect every sensor output, one by one, directly to the JTAG standard interfacing. Then, error location and timing stamp information are forwarded to a dedicated analytics block (A-Block) which, based on machine learning model, performs on-chip analytics. The A-Block outputs are forwarded to a Control Block (C-Block), which is responsible to perform SLM for the SoC.

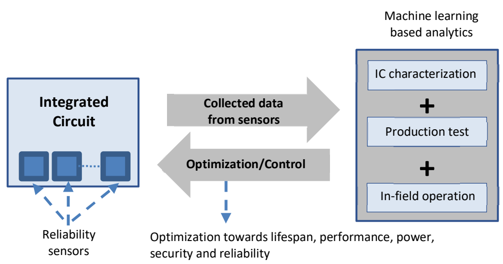

Fig. 5. SLM Platform proposal for space applications.

Fig. 6 depicts a block diagram of the system under consideration. In this scenario, SLM will be carried out by collecting in-field cross layer data from on-chip sensors, from silicon to operating system level. The best machine learning model should be selected from the five candidates depicted in Section 2.2 (B). Our first hypothesis is that the RNN with LSTM model will be the one that will present the best analytics to handle the high complexity and large amount of data collected from the sensors, which must be correlated at different abstraction layers. This is a supervised (off-line) learning model that will be trained based on system information collected during *IC characterization* (e.g., reliability, stress and wear-out tests) and *production test* (e.g., noise immunity, signal time slack in delay-critical paths, minimum operating voltage and I_{DDq}). Once this process is finished, the trained model is mapped to hardware. Furthermore, an unsupervised (on-line) learning model will be used during *in-field operation*. Commonly used on-line algorithms being considered are, among others, the incremental Stochastic Gradient Descent (SGD), as previously mentioned, the Passive-Aggressive and Perceptron models.

For the sake of illustrating a hypothetical system operating as described, assume the following example:
Based on *high* temperature information, *high* conducted-electromagnetic noise on power supply lines, *large* number of SEUs occurring in the system main memory and in delay-critical paths of the ALU, and *increasing* number of faults due to task scheduling disfunction, the A-Block could, based on one or more specific machine learning models, recommend the increase of the frequency by which the EDAC Block washes the memory to counteract with the increasing SEU rate, and reduce the system master clock to try to increase the immunity of the multiplier block to power-supply noise. This countermeasure also minimizes processor power consumption, and attempts to maintain the overall system power consumption as constant as possible, since increasing the EDAC Block wash frequency substantially increases memory power consumption. If these measures are not sufficient, hardware redundancy in the form of functional unit duplication & comparison could be triggered for information stored in memory and timing redundancy could be applied to the multiplier block, for instance. Once the harmful effects from the environment fade away, the machine learning model could recommend the relaxation of these measures so that to minimize power consumption and extend system lifespan. Nevertheless, note that even though the suggested solutions have in mind guaranteeing the system desirable reliability level, these solutions, on the other hand, may increase power consumption, reduces battery lifetime, probably increases temperature and accelerates aging, and ultimately, reduces overall system lifespan at long term. So, there is a trade-off to be considered by the C-Block before taking the final decision to control the system. This approach is what we call by SLM based on in-field cross-layer information.

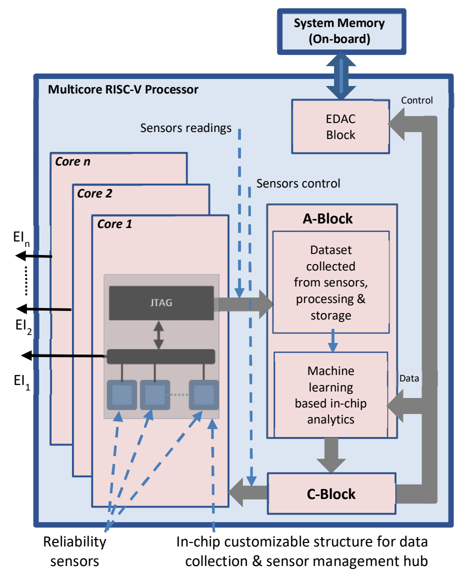

Fig. 6. SLM-based embedded systems for space applications.

Other research topics under consideration are:

a) apply approximate computing in the machine learning model, and then, check if the analytics yields accurate outputs within an acceptable variance. In the future, we expect to be able to apply approximate computing as an effective approach to dynamically trade system power (and so, system lifespan extent) versus reliability. This topic is paramount for battery-powered systems used in space applications.

b) apply rejuvenation algorithm to a functional unit when needed. After aging sensors indicate that a given data-path is aged in excess, A-Block could recommend the execution of a rejuvenation algorithm in order to partially recover that functional unit. According to [18], [19] a given path ages as function of a combination of three factors: (a) as function of the logic state of the nodes along the path (which is directly dependent on the applied inputs to the circuit); (b) as function of the operating temperature, and (c) as function of the frequency by which the path nodes change logic state. In more detail, depending on the logic state of the inputs of a gate, the transistors could be found in the *stress* or in the *recovery* state. Moreover, *static* inputs are the worst-case for aging a path, since even though some of the gates along the path will be in the *recovery* mode, others will be in the *stress* one. If some gates along the path suffer from excessive delay, there is a large probability that timing faults will be produced at the path output. As observed by [18], [19], this fact conducts to the conclusion that a path is less exposed to aging when its nodes are always changing the logic state (as faster as possible). The same reasoning is true for SRAM cells. When a cell is storing the same logic value for long time periods, aging is more aggressive than when the cell is being successively written with the opposed value (this behavior is also dependent on the frequency by which the cell is being written, as higher the frequency, smaller the aging effect on that cell). On the contrary, if the cell is storing the same logic value for long periods, aging accelerates and the SRAM cell, at advanced aging states, behaves as a ROM memory. In other words, the cell can only be read, or be written with the same logic state. If the cell is written with the opposite value, it behaves like in the presence of a stuck-at on the value stored previously to the write operation. In other work, [20] authors demonstrate that SRAM cells in a commercial FPGA also age to ionizing radiation in the same way (authors called this phenomenon the "imprint effect"), i.e., when fresh, the cells operate as SRAM, and when advanced aging states dominate, the cells operate as ROM. In this scenario, rejuvenation algorithms can be applied to logic as well as to large SRAM blocks. To do so, the C-Block can interrupt the processor, forcing it to execute a specific rejuvenation procedure in logic and/or memory. In fact, designers are very concerned about aging of SRAM IP blocks devoted to space systems because, as pointed by [21], in addition to shifting functionality from SRAM to ROM behavior, aged cells present higher SEU rate in comparison to fresh, non-aged cells. This is observed because aged SRAM cells, before operating as ROM, become unbalanced, and so, can easily flip from one state to another, due to a particle strike.

III. Conclusions

This paper discussed the high complexity of designing resilient systems for space applications. There is an endless number of parameters that can be combined in different degrees to induce system failure: temperature, conducted-electromagnetic noise on power-supply lines, power-supply fluctuation due to battery disruption, aging due to cumulated radiation (TID), aging due to natural wear-out (HCI, BTI), transient faults (SEU, SET), electromigration on IC tracks experiencing large current densities, and unavoidable process variations.

In this challenging scenario, IHP is tackling the complexity of designing space systems with on-chip capability for SLM. Such design is based on extensive cross-layer information collection, from silicon to operating system, in order to feed large amounts of data to on-chip analytics. It is our expectation that the analytics outputs will yield accurate enough directives to allow high-quality SLM. With this purpose, IHP has adopted the RISC-V open-source instruction set architecture (ISA) as the baseline computation platform running a customized version of the FreeRTOS operating system. The platform is designed with the IHP 130 nm SiGe BiCMOS rad-hard technology, which is currently under evaluation for inclusion in the European Preferred Parts List (EPPL) of the European Space Agency (ESA).

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